

CS & IT ENGINEERING



Digital Logic

Optimisation

Lecture No. 06



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Recap of Previous Lecture



Topic



K-maps (SOP)

Topics to be Covered



Topic



{ Practice K-maps }

{ K-maps (POS) }

☆ Advance K-maps

(Q)

$$f(P, Q, R, S) = \sum(0, 1, 2, 4, 6, 9, 10, 11, 12, 13, 15)$$

pair $\rightarrow$ 3 var
Quad $\rightarrow$ 2 var

~~Lessen 90p
Largen 90p~~

$PQ \diagdown$	$RS \rightarrow$				
	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$	
$\bar{P}\bar{Q}$	1 0	1 1			3
$\bar{P}Q$	1 4			1 6	7
$P\bar{Q}$	1 12	1 13	1 15		14
PQ		1 9	1 11	1 10	

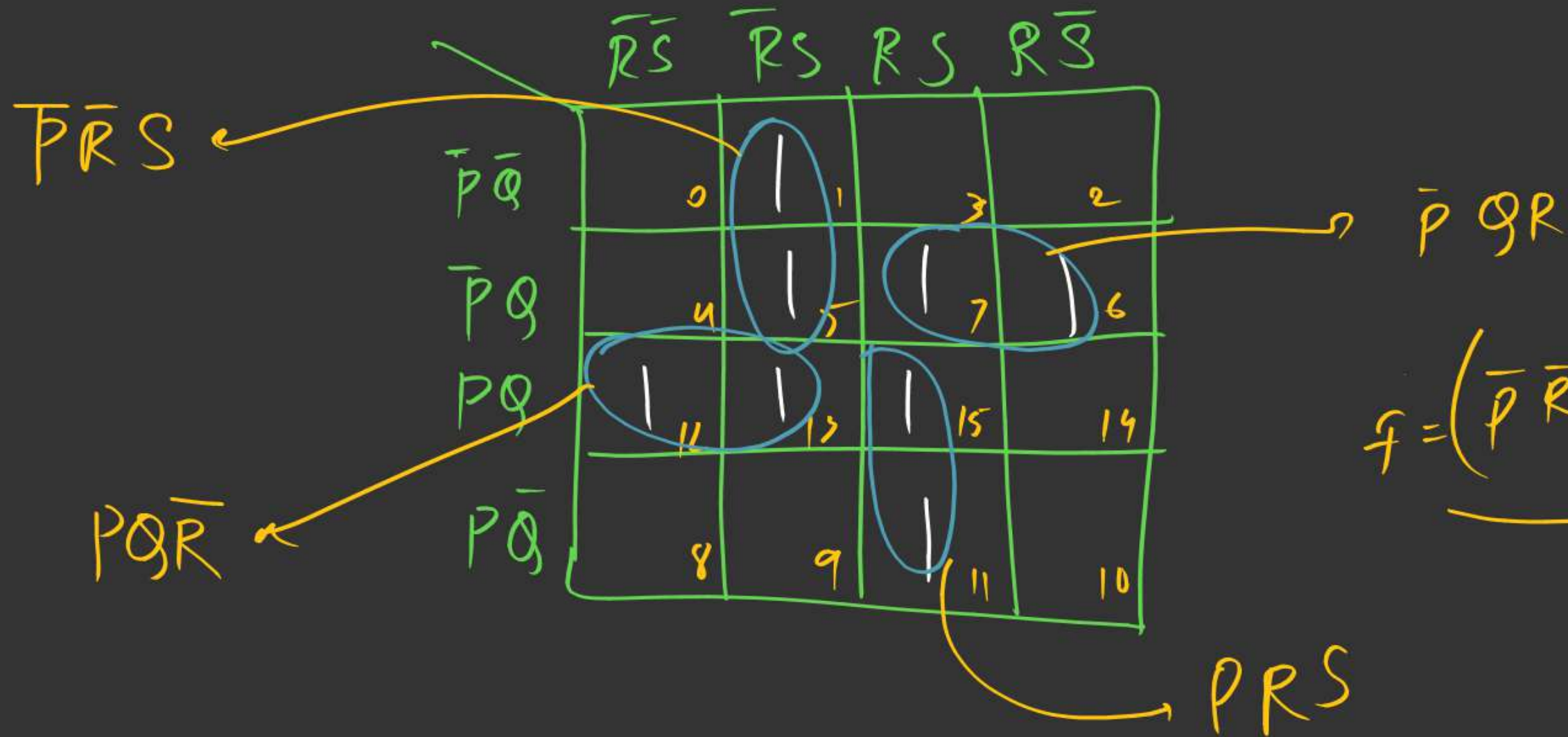
2 Quad
3 pairs

1 Quad
4 Pairs

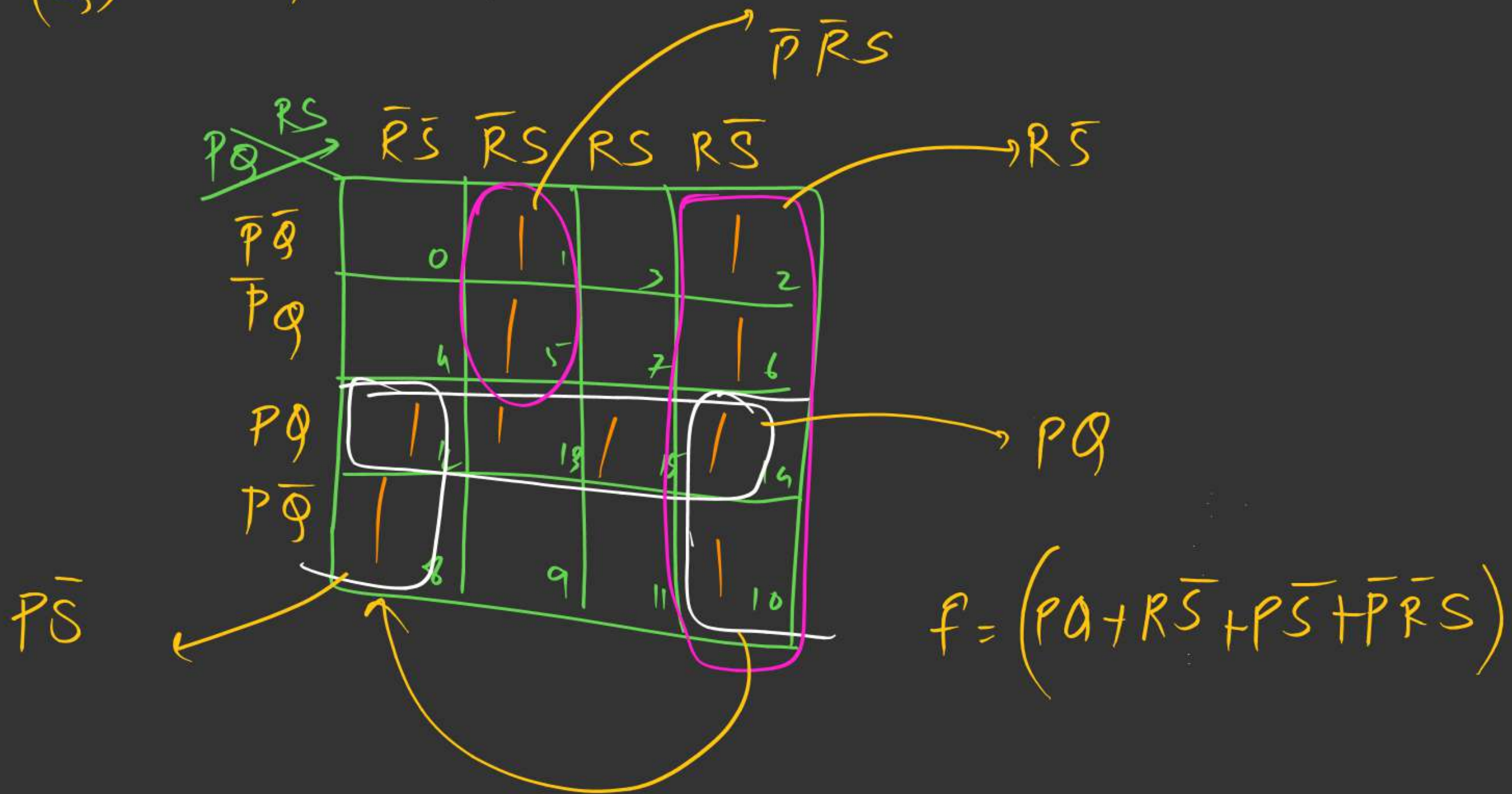
V. Imp

HW:-

$$f(P, Q, R, S) = \sum (1, 5, 6, 7, 11, 12, 13, 15)$$



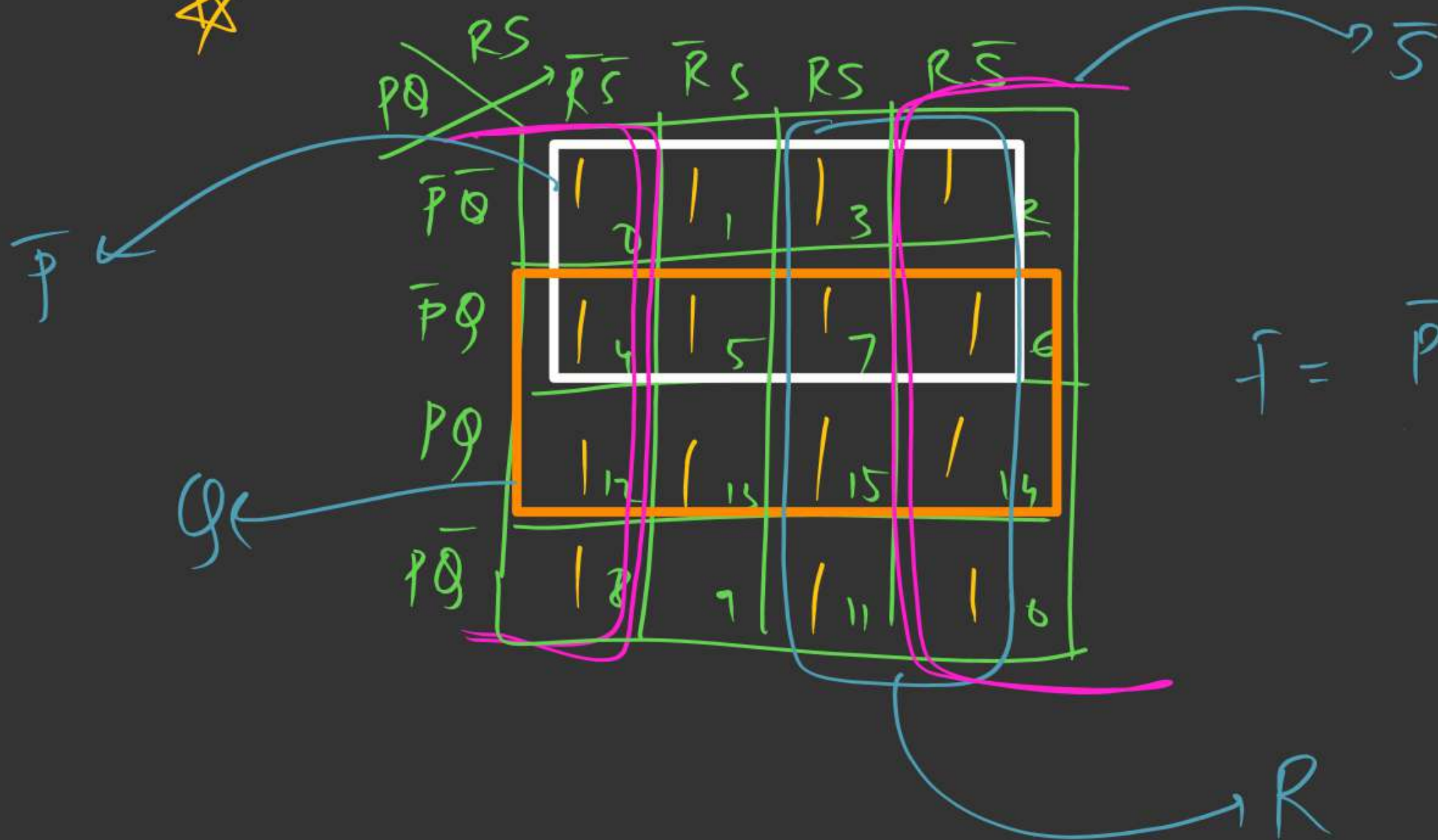
(Q) $F(P, Q, R, S) = \sum (1, 2, 5, 6, 8, 10, 12, 13, 14, 15)$



V-Imp

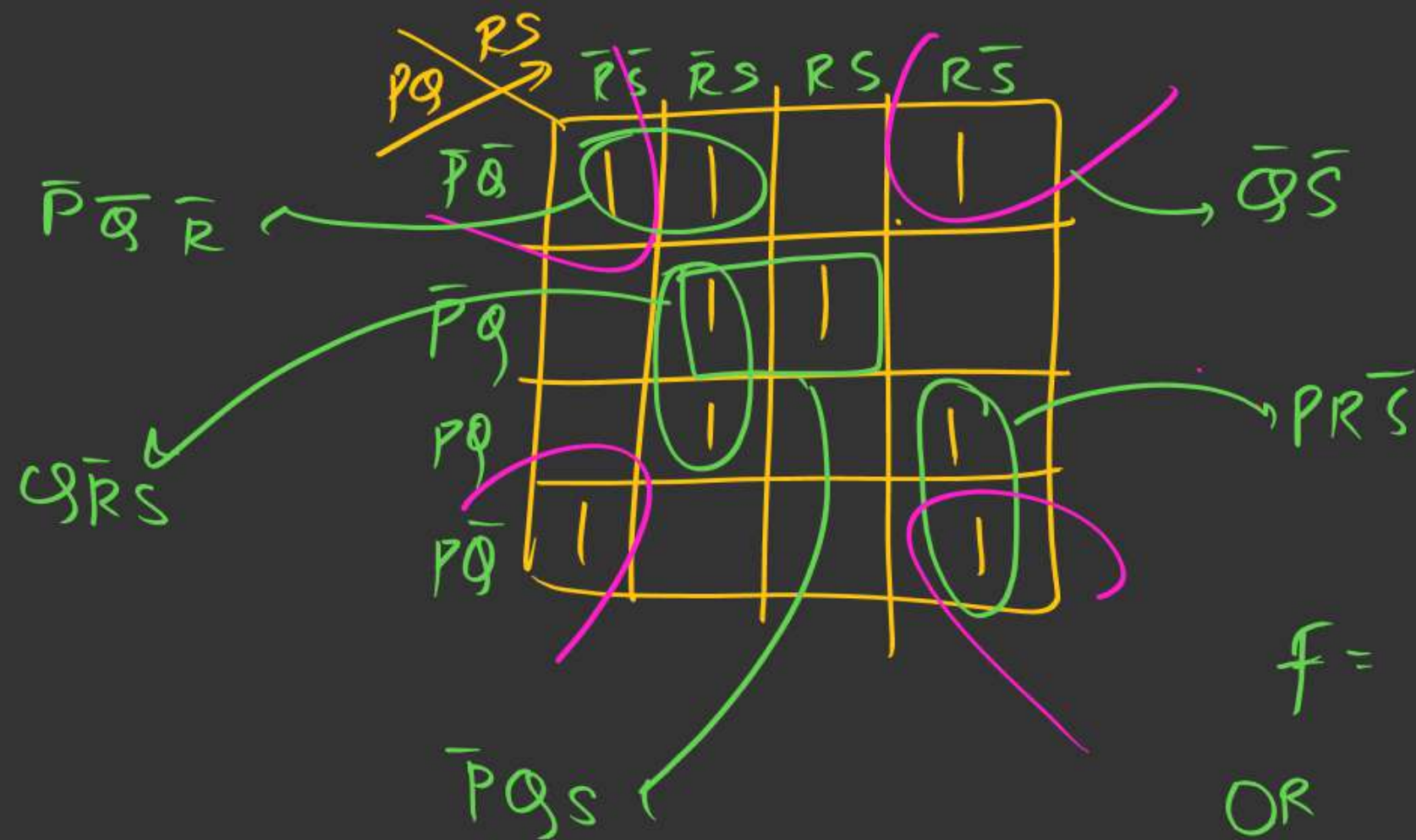
(Q) $f(P, Q, R, S) = \Sigma(0, 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15)$

★



$$f = \bar{P} + Q + \bar{S} + R$$

$$(Q) \quad f(P, Q, R, S) =$$



$$f = \bar{Q}\bar{S} + PR\bar{S} + \bar{P}QS + Q\bar{R}S + \bar{P}\bar{Q}R$$

OR

$$f = \bar{Q}\bar{S} + PR\bar{S} + \bar{P}QS + Q\bar{R}S + \bar{P}\bar{Q}R$$

Don't Care Conditions:

↳ Combinations of i/p on which the o/p may or may not depend.

$$F(P, Q) = \bar{P}Q + QP + \textcircled{\bar{Q}\bar{P}}$$

optional
↓
don't care

$$(Q1) \quad F(P, Q, R) = \sum m(0, 1, 6, 7) + \sum d(3)$$

don't care X

↓
don't Care

		$\overline{Q}\overline{R}$		$\overline{Q}R$		QR		$Q\overline{R}$	
		00	01	11	10				
P	$\overline{Q}$	1	1	X					
	Q								
	0								
	1								

$\overline{P}\overline{Q}$ ← (points to the first row of the K-map)
 PQ ← (points to the last row of the K-map)

$$\underline{\underline{F = \overline{P}\overline{Q} + PQ}}$$

$$(Q.2) \quad f(P, Q, R) = \sum m(0, 1, 2) + \sum d(3, 6)$$

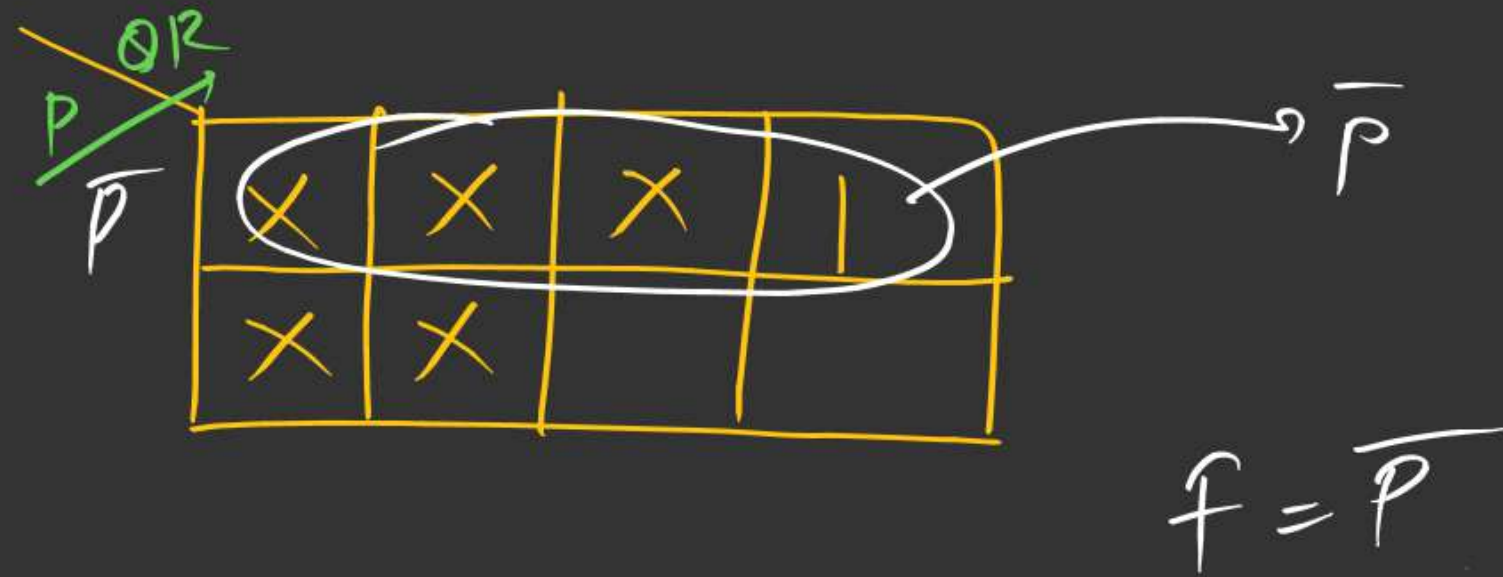
P \ QR				
	$\overline{Q}\overline{R}$	$\overline{Q}R$	QR	$Q\overline{R}$
$\overline{P}$	1 0	1 1	X 3	1 2
P				X 6
	4	5	7	6

$$f = \overline{P}$$

	$\bar{Q}\bar{P}$	$\bar{Q}R$	QR	$Q\bar{R}$
$\bar{P}$	X ₀	1 ₁	1 ₃	2
P	4	X ₅	1 ₇	X ₆

R

$$\bar{F} = R$$



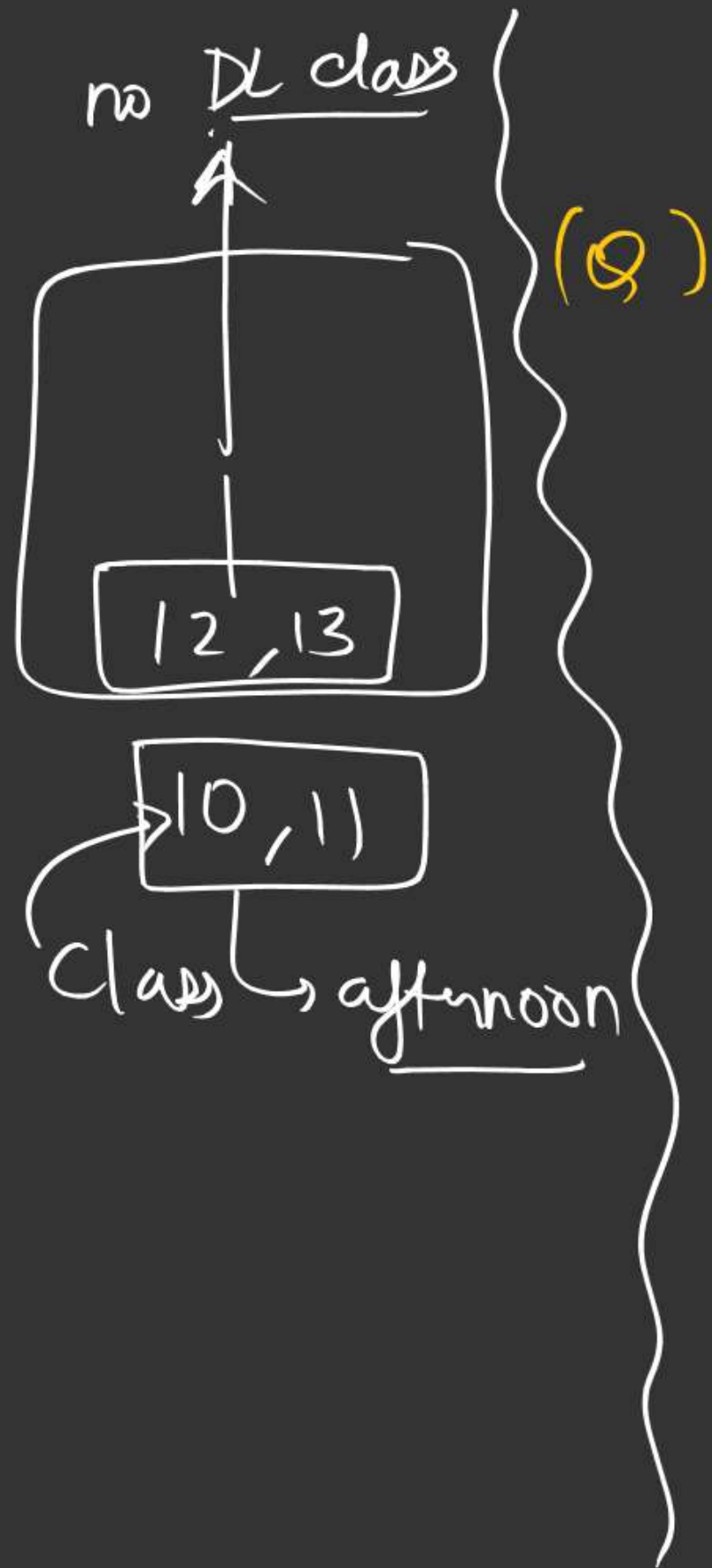
$$(Q) F(P, Q, R, S) = \sum m(0, 2, 4, 6, 7, 8, 10, 11, 12, 14, 15)$$

$$+ \sum d(1, 3)$$

$PQ \backslash RS$	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	1 0	X 1	X 3	1 2
$\bar{P}Q$	1 4	5	1 7	1 6
$P\bar{Q}$	1 12	13	1 15	1 14
PQ	1 8	9	1 11	1 10

$$f = (R + \bar{S})$$

R



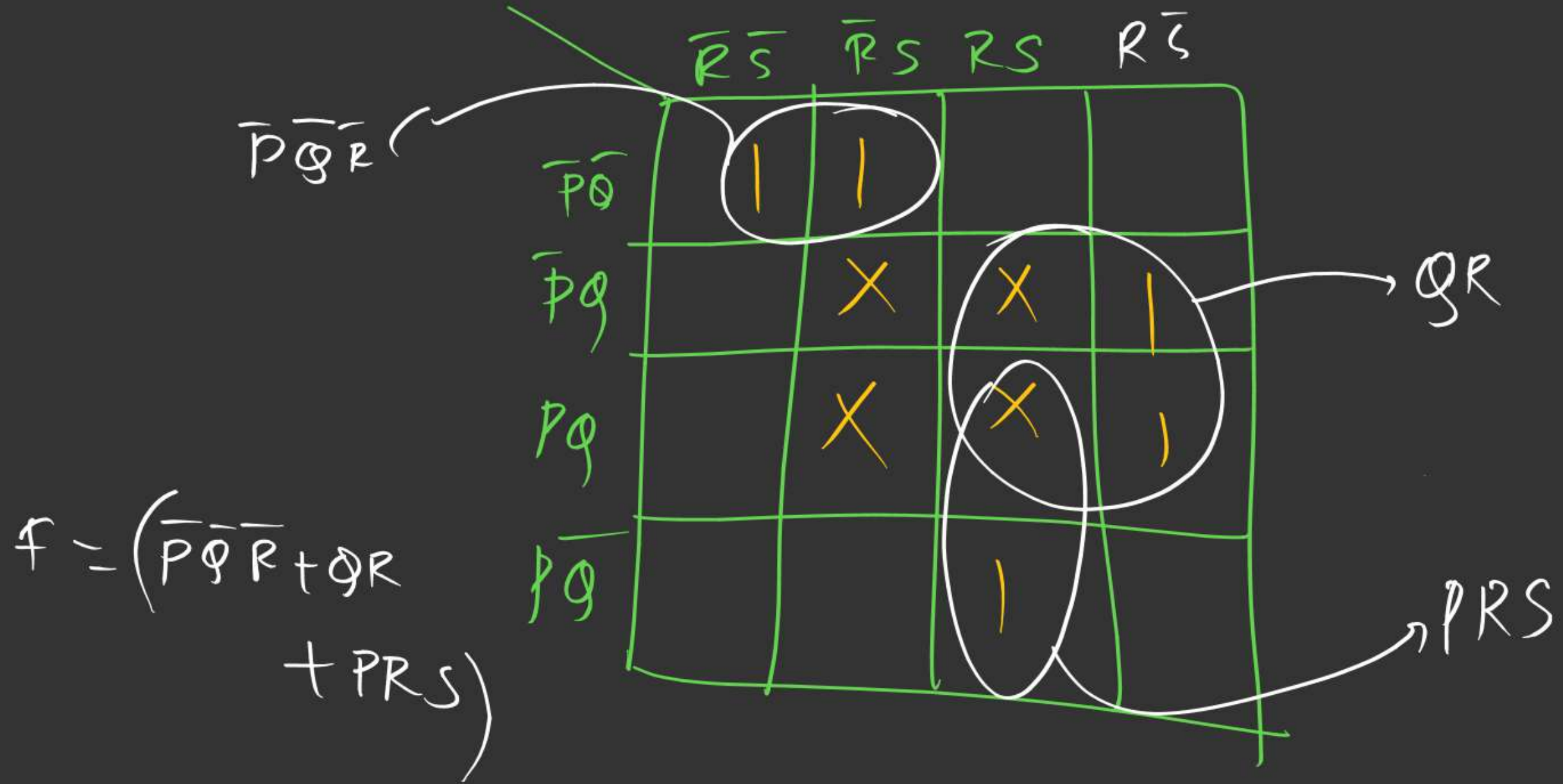
$\bar{P}\bar{Q}$

$\bar{Q}\bar{S}$

QRS

	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	1 0	1 1	X 3	1 2
$\bar{P}Q$	4	5	1 7	X 6
$P\bar{Q}$	12	X 13	1 15	14
PQ	X 8	X 9		1 10

$$F = (\bar{P}\bar{Q} + \bar{Q}\bar{S} + QRS)$$



k-maps

AB

$$00 \rightarrow (A+B)$$

$$10 \rightarrow (\bar{A}+B)$$

POS



$$0 \rightarrow A$$

$$1 \rightarrow \bar{A}$$

$$(Q) = f(P, Q, R) = \prod M(0, 1, 3, 6, 7) \\ = \sum m(2, 4, 5)$$

mtd-1

SOP-Kmap

Bonus

90p → 0

P \ QR	$\bar{Q}\bar{R}$	$\bar{Q}R$	QR	$Q\bar{R}$
$\bar{P}$	0 ₀	0 ₁	0 ₃	1 ₂
P	1 ₄	1 ₅	0 ₇	0 ₆

$\bar{P}\bar{Q}$

PQ

$$F = \bar{P}\bar{Q} + PQ + \bar{P}R$$

$$F = \frac{\bar{P}\bar{Q} + PQ + \bar{P}R}{\bar{P}\bar{Q} + PQ + \bar{P}R}$$

POS

$$= \overline{\bar{P}\bar{Q}} \cdot \overline{PQ} \cdot \overline{\bar{P}R}$$

$$= (P+Q) \cdot (\bar{P}+\bar{Q}) \cdot (P+\bar{R})$$

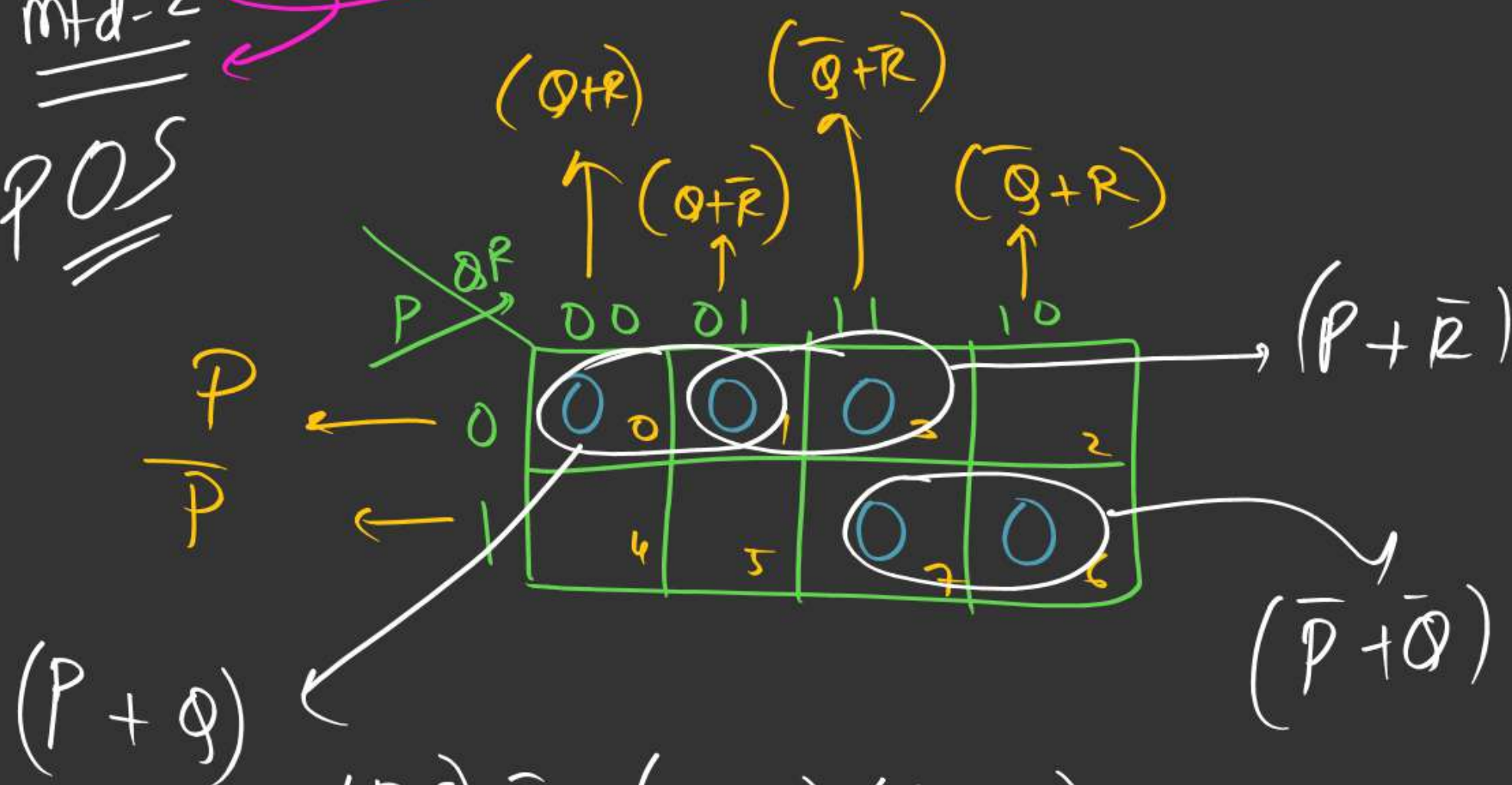
→ Ans (POS)

not Recommended

Standard
 m+d-2
POS

use this

$$F(P, Q, R) = \Pi M(0, 1, 3, 6, 7)$$



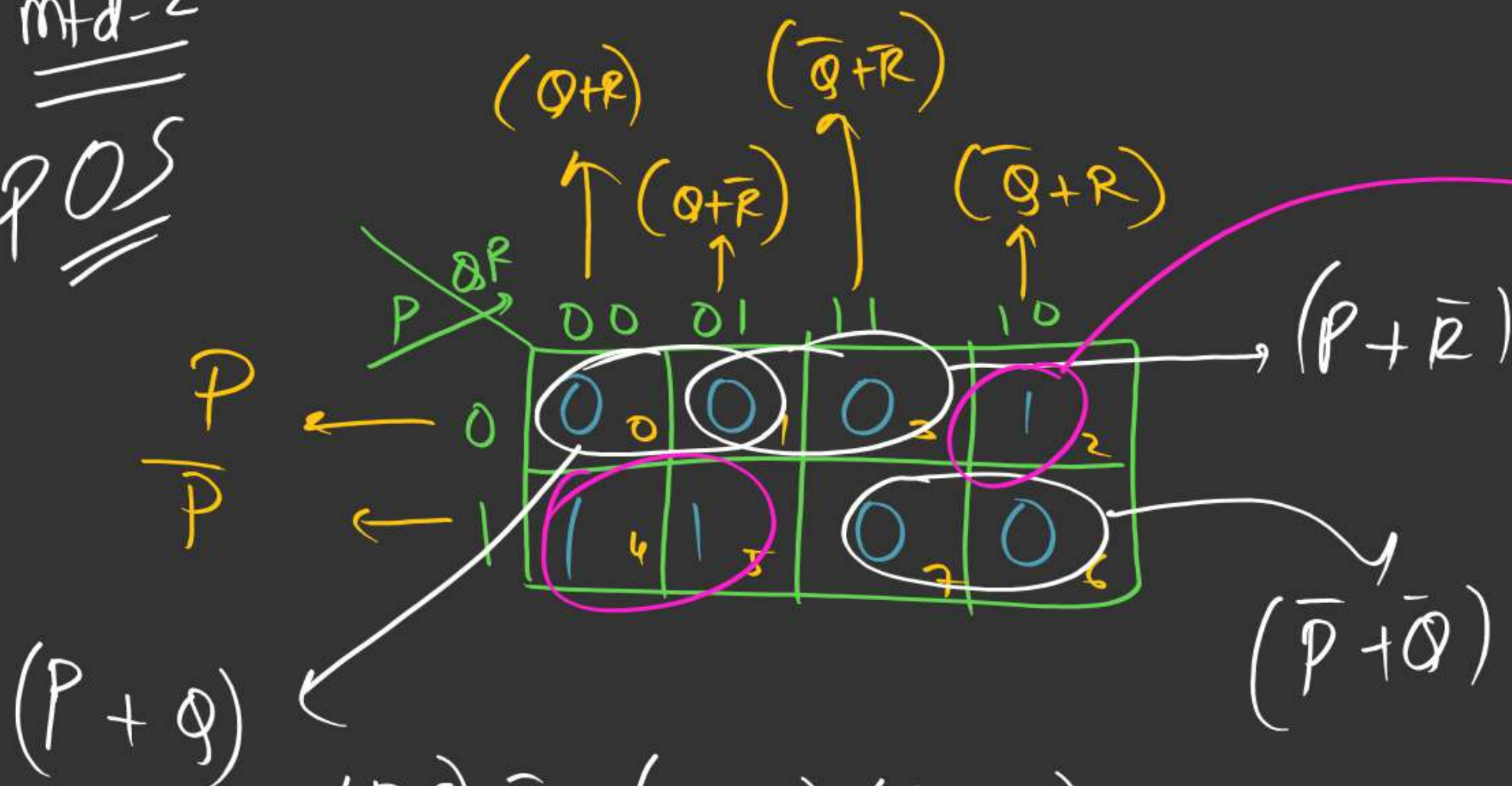
$$\underline{\underline{(POS)}} \quad F = (P + Q)(\bar{P} + \bar{Q})(P + \bar{R}) \longrightarrow \underline{\underline{Ans}}$$

Standard

$$F(P, Q, R) = \Pi M(0, 1, 3, 6, 7)$$

m+d-2

POS



SOP $\Rightarrow$
 $(P\bar{Q} + \bar{P}Q\bar{R})$

(POS) $F = (P+Q)(\bar{P}+\bar{Q})(P+\bar{R}) \rightarrow$ Ans

$$F(P, Q, R, S) = \prod M(2, 4, 5, 8, 9, 12, 13, 14, 15)$$

POS

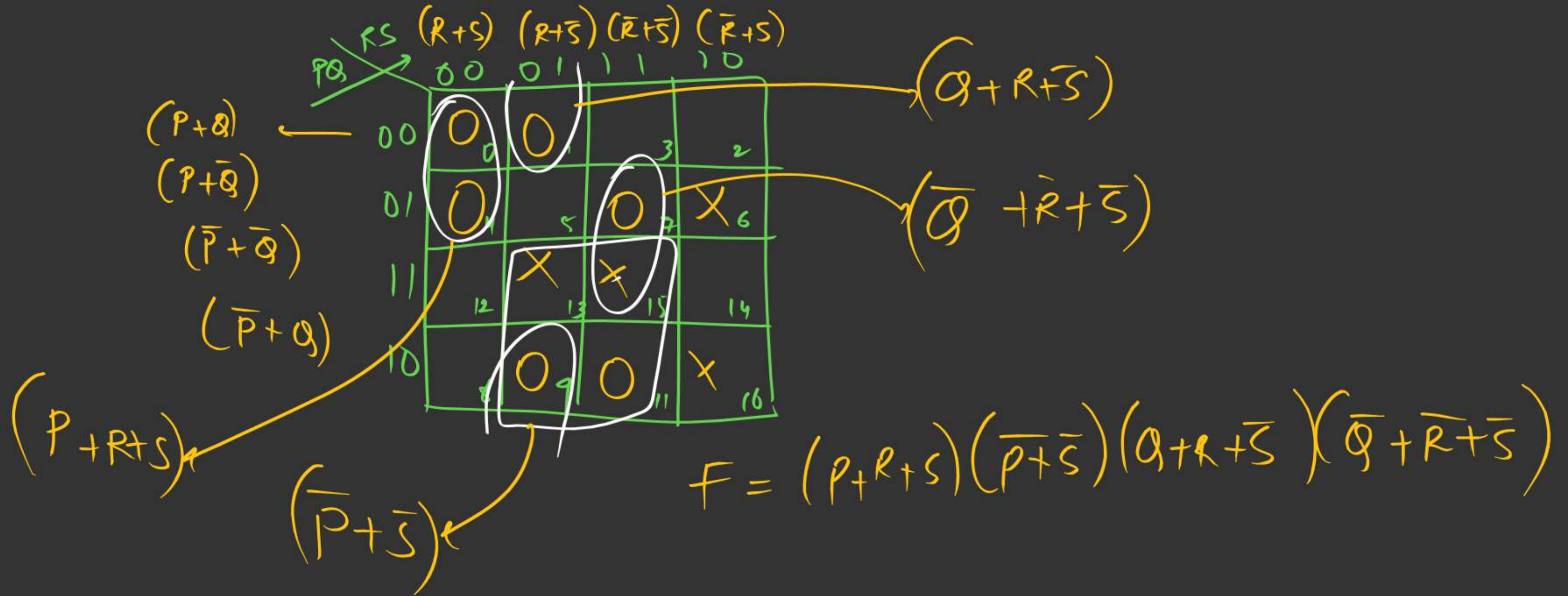
		$(R+S) \quad (R+\bar{S}) \quad (\bar{R}+\bar{S}) \quad (\bar{R}+S)$			
		00	01	11	10
$PQ \backslash RS$					
$(P+Q) \leftarrow 00$	00	0	1		2
$(P+\bar{Q}) \leftarrow 01$	01	4	5	7	6
$(\bar{P}+\bar{Q}) \leftarrow 11$	11	12	13	15	14
$(\bar{P}+Q) \leftarrow 10$	10	8	9	11	10

$(\bar{P}+R)$ (points to cells 8, 9, 12, 13)
 $(\bar{Q}+R)$ (points to cells 5, 7, 13, 15)
 $(\bar{P}+\bar{Q})$ (points to cells 12, 13, 14, 15)
 $(P+Q+\bar{R}+S)$ (points to cell 2)

POS: $(\bar{P}+R)(\bar{Q}+R)(\bar{P}+\bar{Q})(P+Q+\bar{R}+S)$

pos + dont care:-

$$F(P, Q, R, S) = \prod M(0, 1, 4, 7, 9, 11) + \prod d(6, 10, 13, 15)$$



HW1

$$(Q.1) F(P, Q, R, S) = \sum m(0, 2, 4, 6, 7, 8, 10, 11, 12, 14, 15)$$



- 1) SOP
- 2) POS



2 mins Summary



k-maps



THANK - YOU

